Binary Classification of AOI Alarms into True Defects and False Calls

Second check of AOI results with supervised machine learning (Siemens production data)

P2: core code and baseline simulation · test plan · report

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Team 1 · Guided by Prof. Dr.-Ing. Hamidreza Heidari · 3 November 2026 · github.com/bodhi584/aoi-false-call-classification

Yu Han

Where the project stands

P1	20 Oct	Proposal, tech stack, MVP (revised topic 23 Oct)	**done**
P2	3 Nov	Core code + baseline simulation; test plan; report	**this phase**
P3	24 Nov	Feature progress; unit tests and CI; issues board	**next**
P4	8 Dec	Feature progress; full evaluation	**planned**
P5	22 Dec	Demo video, run instructions, packaging	**planned**
Final	12 Jan	Final presentation and report	**planned**

P2 in short

(summary of this presentation, written last)

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Melisa Subasi

AOI: what it is, how it works, why false calls matter now

≈ 99 %

of the solder joints that AOI flags are false calls [8]

Every flagged board is checked again by a person. Siemens line, 132 days, 440,274 flagged joints [5].

1 · What AOI is and where it is used

Review paper [2] · Abu Ebayyeh & Mousavi, IEEE Access 8, 2020

In electronics assembly (SMT), a camera system inspects every solder joint after soldering: it measures the joint and compares the measurements with fixed limits.

open paper (doi)

2 · How it works on a real line, and the problem

Data article + open data set [5] · Pfab et al., Data in Brief 53, 2024 · Siemens SMT line

If one measurement is outside its limit, the board goes to a manual inspection station; an operator decides: true defect or false call. Almost all alarms are false calls, so operators mostly check good boards.

open paper (doi)

3 · Why it matters now

Press release [4] · Siemens and NVIDIA, Erlangen electronics factory, 21 Oct 2026

Siemens is adding a second inspection step that decides whether an AOI alarm is a real defect or a false call; target 40 % fewer false calls (a target, not a result).

open press release

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Second check with machine learning: three studies, one baseline

Study	Type	Data	Classifiers	Later periods?	Code
[6] Thielen et al. 2020	conference (ISSE, IEEE)	own, 1,144 / 4,264 components, not public	kNN, neural network, random forest	not tested; threshold tuned on the evaluation data	no
[7] Albayari et al. 2026	conference (FAIM 2025, Springer)	Siemens AG; not stated whether the public set	XGBoost, RF, DNN, kNN	not tested; several train–test splits	no
[8] Pfab & Rothering 2025	preprint (arXiv)	Siemens, public [5]	kNN, RF, balanced RF, XGBoost, AutoML	yes: 5 later slices, 5–15 % missed defects	yes, with run logs

What all three show

A classifier on AOI measurements recognises false calls on a test set from the same period; random forests are among the best models [6], [8].

Why [8] is our core paper

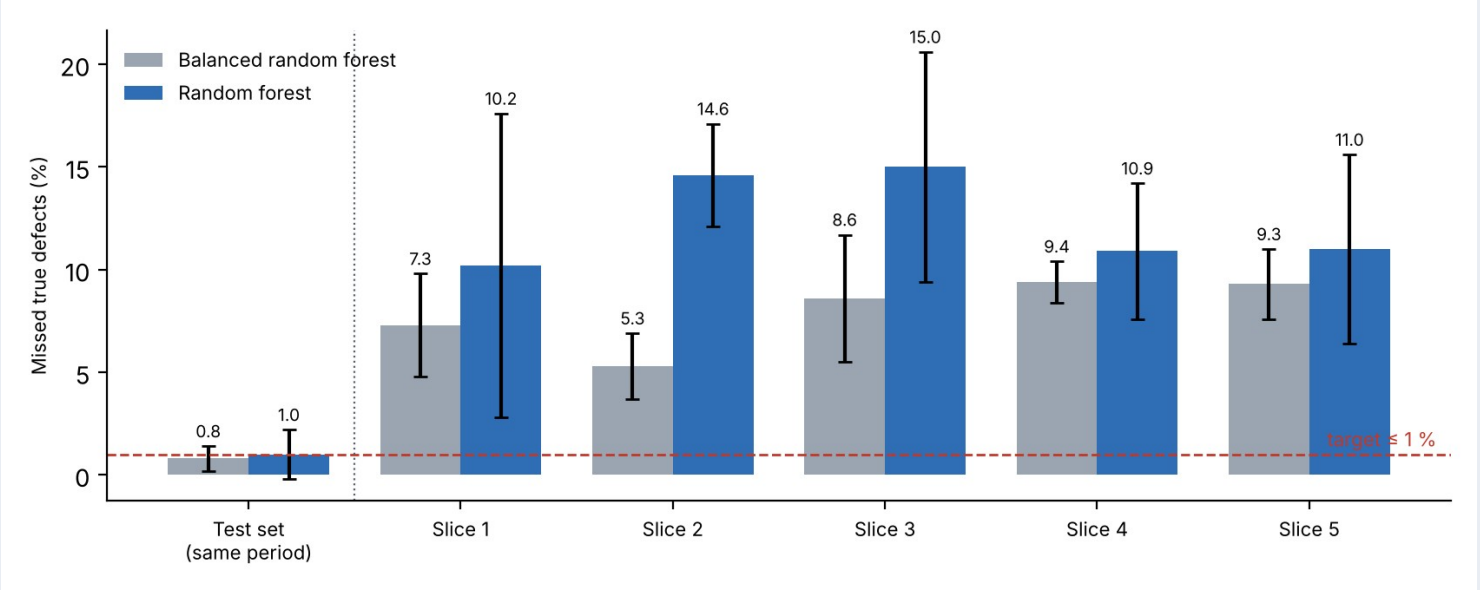
Same public data as ours, published code and run logs, and the only study that tests later production periods. We can repeat it exactly and build on it.

[6] doi [7] doi [8] arXiv [8] code

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[8] in detail: works on the test set, fails on later production days



Two causes named in [8]

Trained once on the first half; on target on the test set; 5–15 % missed defects on all five later slices.

Drift = the data change over time; a model trained on old data no longer fits [21]–[23].

Same effect in manufacturing: welding, semiconductor etch, sensors [24]–[26].

Threshold: the paper names its own threshold method as unstable [8], [28].

Both named as “potential next steps”, neither tested.

Meron Hagos

Chain of evidence

Fails over time

train once → 5–15 % missed defects later [8]

Two causes

drift [21]–[27] and an unstable threshold method [8], [28]

Methods exist

threshold: [28]–[30]; retraining: [22], [27], [31]–[33]

Works elsewhere

fraud detection, <1 % positives [35]; semiconductor materials line [36]

Open for AOI

named by [8] as next steps, not tested for the second check of AOI results

Pfab & Rothering (2025), Sec. VII: "… potential next steps to stabilize the modeling by improving the method for setting the threshold and handling the distribution drifts within the dataset."

Our simulation focuses on exactly these two steps: the threshold method and drift (retraining).

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Meron Hagos

The two next steps of [8] become the variables of our simulation

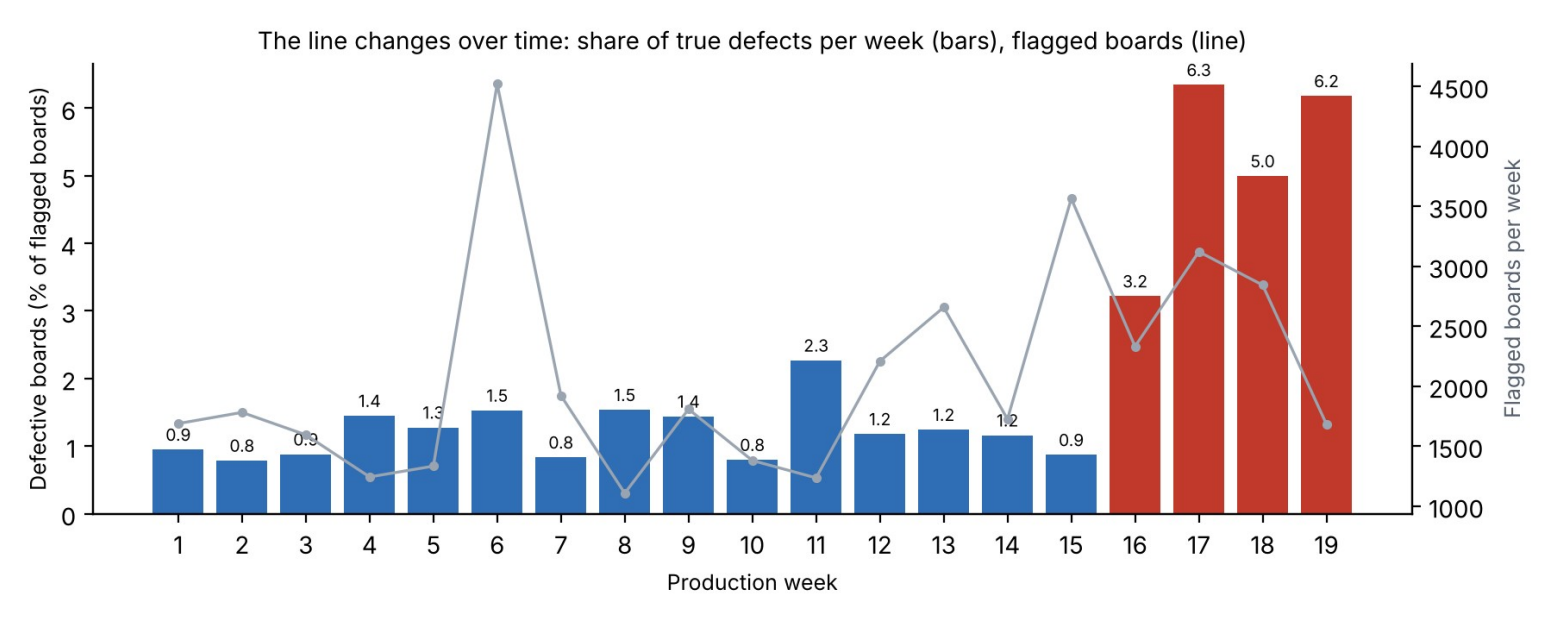
Variable	Option	What it does	Sources
Threshold	as in [8]	mean of the five cross-validation thresholds	[8]
	order statistic with confidence	1 % bound held with 95 % confidence; needs ≥ 299 left-out true defects	[28]
	re-set for the current period	follows a changed share of true defects, also without labels	[29], [30]
Retraining	off	train once, as in [8]	[8]
	fixed schedule	per period, weekly, daily; growing or sliding window	[22], [31], [35], [36]
	triggered by drift	retrain when the error rises, or adapt the window	[32], [33]

Every option comes from a published method; the simulation shows its effect on missed defects and released false calls next to [8].

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Meron Hagos

The line changes over time



Facts (our analysis)

440,274 flagged joints, 39,742 boards, 132 days

97.8 % of flagged boards are false calls

Share of defective boards: 0.8–2.3 % per week, then 3.2–6.3 % in the last 4 weeks

Drift test: early vs. late days separable, AUROC 0.974

Alejandro Rivas

Input → simulation → output

INPUT

Recorded Siemens production

AOI measurements of each flagged solder joint, in time order [5]

published models and settings of [8]

VARIABLES

What operation would change

Retraining: off / on, frequency (per slice → weekly → daily)

Threshold: as in [8] / set correctly / set on recent days

OUTPUT

Per production period

Missed true defects (%)

False calls released (%) = manual checks saved

next to the result of [8]

Reference lines from [8]: ≤ 1 % missed true defects, ≥ 40 % false calls released.
Later (P3–P5): a software with these variables as settings and charts of both outputs.

Alejandro Rivas

Replication of Pfab & Rothering (2025): one factor at a time

	(a) Paper exactly	(b) Train once, threshold re-estimated	(c) Retrain
Data split	same as [8]	same	same
Models, hyper-parameters, seeds	RF, balanced RF; logged values; 10 seeds	same	same
Threshold	value logged in [8]	their method, applied to the final model	same method, new data
Training	once	once	before each of the 5 later slices
Purpose	reproduce [8]	(a) vs (b) = effect of the threshold	(b) vs (c) = effect of retraining

Metrics on joint level, as in [8]: missed true defects (slip = 1 − recall) and false calls released (volume reduction = specificity). Assumption: manual-inspection labels are available for all joints (upper bound for retraining).

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What runs in the repository

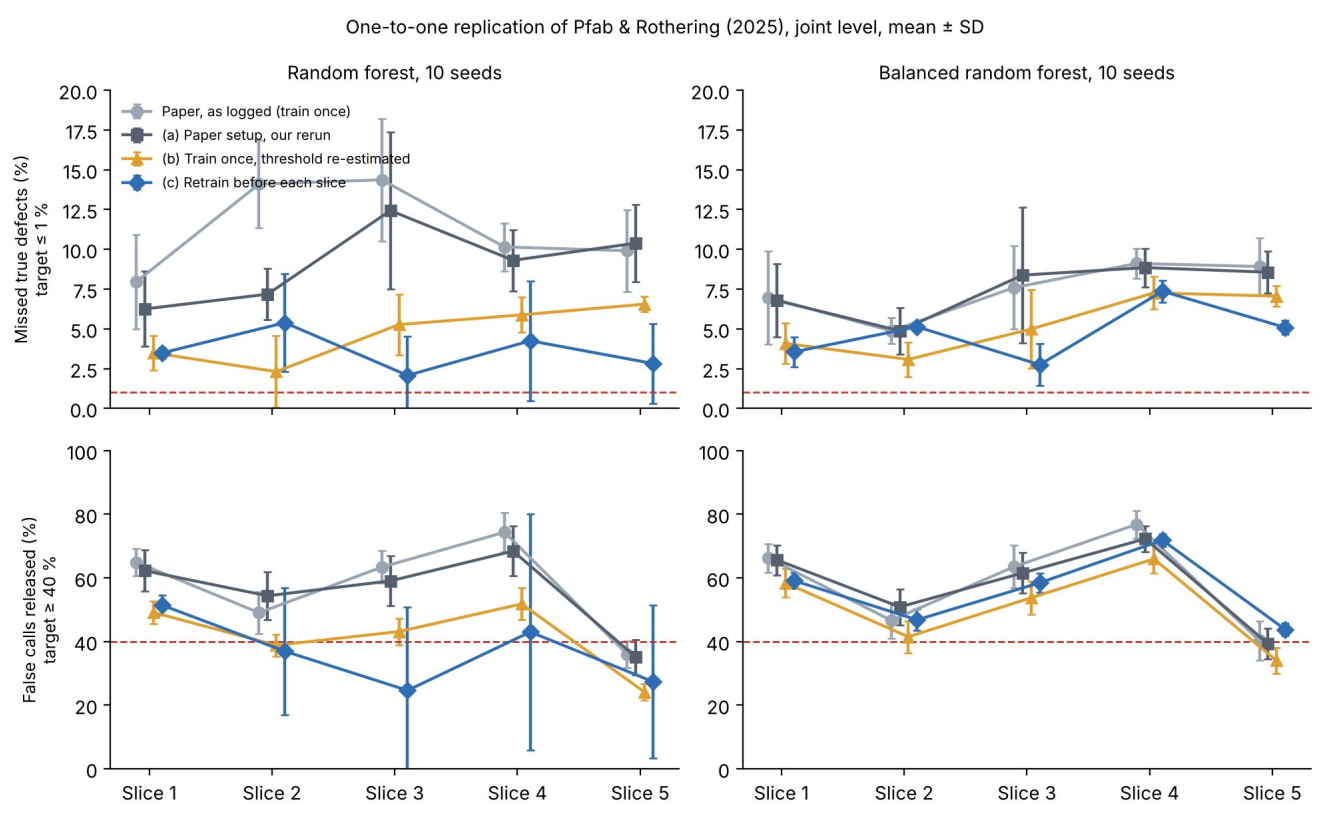
Part	File	What it does
Data	aoi_classification/data.py	load the Siemens data, boards, time order
Classifier + threshold	aoi_classification/model.py	train, score, threshold for 99 % recall
Baseline simulation	aoi_classification/simulation.py	day by day: train on earlier days only
Replication	scripts/replicate_pfab2025.py	one-to-one repeat of [8], train once vs. retrain
Classifier screening	scripts/compare_classifiers.py	13 classifiers, 8 families, same split
Tests + CI	tests/, GitHub Actions	10 tests on Windows, macOS, Linux

Tools: Python, pandas, scikit-learn, imbalanced-learn, matplotlib, Jupyter · Data: Mendeley (CC BY-NC 3.0), not stored in the repo

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Baseline reproduced; threshold and retraining both lower missed defects



Reproduction ✔

Balanced RF: 7.5 % missed defects, as in [8].

Retraining: small gain

Balanced RF: 5.3 → 4.8 % missed, 50.7 → 56.0 % released.

RF: 4.7 → 3.6 % missed, but fewer released.

Threshold matters most

Correct threshold alone: 7.5–9.1 → 4.7–5.3 % missed.

Reference line ≤ 1 % not reached yet.

Yu Han

How we check the code and the numbers

Group	Examples	Status
Data (D1–D7)	labels 0/1, no missing values, time order, checksum	4 exist, rest P3
Classifier + threshold (C1–C7)	threshold rule, all models run, same seed = same result	4 exist, rest P3
Simulation (S1–S4)	model only knows earlier days	1 exists, rest P3
Replication (R1–R4)	same split as [8]; (a) reproduces [8]; retraining uses only earlier data; 10 seeds	new
KPIs (K1–K5)	missed defects ≤ 1 %, false calls released ≥ 40 %; every number from results/	open

Unit tests run on every push (CI, three operating systems). A KPI that is not reached is a result, not a failed test.

Yu Han

Revision of the P1 proposal after finding the core paper [8]

	What [8] already did	P1 plan (23 Oct)	Revised plan (P2)
Second check	classifier after AOI on the Siemens data; on a same-period test set ≤ 1 % missed defects	build our own second check	use the check of [8] as baseline (one-to-one)
Over time	evaluated on 5 later slices with a static model: 5–15 % missed defects	day-by-day simulation in time order	day-by-day simulation of operation on top of [8]: the part they did not do
Causes	names drift and the threshold method; tests neither	not known yet	address both: retraining against drift; a correctly set threshold
Models	random forest, balanced random forest (also kNN, XGBoost, AutoML)	logistic regression, random forest, neural network	same as [8], so results are comparable; model choice matters little [13]
Comparison	targets ≤ 1 % missed, ≥ 40 % false calls released	recall ≥ 99 %, specificity ≥ 20 % (board level)	same metrics as [8]; compared with their later-slice results

Unchanged: Siemens AOI data [5] · input → system → output · tools (Python, scikit-learn) · milestones

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Towards P3 (24 Nov)

Simulation software

one screen: retraining and threshold as settings; charts of missed defects and false calls released next to [8]

Retraining frequency

daily, every 3 days, weekly, per slice: how far do missed defects drop? [36]

Threshold with confidence

order statistic with enough left-out true defects [28]; re-set for the current period [29], [30]

More model families

gradient boosting, neural network, logistic regression in the same setup

KEY REFERENCES

Full list (36 sources) in docs/literature_report.pdf

[5] Pfab, K. et al. (2024). Data of automated optical inspection of surface-mounted technology electronic production. Data in Brief 53, 110110.

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[28] Tong, X., Feng, Y., Li, J. J. (2018). Neyman-Pearson classification algorithms and NP receiver operating characteristics. Science Advances 4(2), eaao1659.

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Repository

github.com/bodhi584/aoi-false-call-classification

Thank you

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